

Practice3_Barraza_Torres_U2

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```
library(titanic)
data("titanic_train")
```

#CONSTRUIR UNA TABLA DE CONTIGENCIA (SEX AND SURVIVE) Y REALIZAR UNA LIMPIEZA DEL DATA SET

```
info <- subset(titanic_train, select = -c(PassengerId, Pclass, Name, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked))
cleaninfo <- na.omit(info)
```

Cross-tabulation or CrossTab

Cross-tabulation, often shortened to cross tab, is a statistical tool used to examine the relationship between two or more categorical variables. It works by organizing data into a contingency table, which displays the frequency (or count) of observations that fall into each combination of categories.

1.- CALCULAR PROBABILIDAD CONJUNTA DE SER MUJER Y SOBREVIVIR

```
freq_femsur <- nrow(subset(cleaninfo, Sex == "female" & Survived == 1))
total <- nrow(cleaninfo)

prob_Fsur <- freq_femsur/total
cat("The probability of a female surviving is for:", prob_Fsur*100, "%")

## The probability of a female surviving is for: 26.15039 %
```

#PROBABILIDAD CONJUNTA DE SER HOMBRE Y NO SOBREVIVIR

```
freq_malenosur <- nrow(subset(cleaninfo, Sex == "male" & Survived == 0))

prob_Mnsur <- freq_malenosur/total
cat("The probability of a male NOT surviving is for:", prob_Mnsur*100, "%")

## The probability of a male NOT surviving is for: 52.52525 %
```

#CALCULAR PROBABILIDAD CONDICIONAL DE SOBREVIVIR DADO QUE ES MUJER

```

female <- subset(cleaninfo, Sex == "female")
fem_survived <- nrow(subset(female, Survived == 1))
female <- nrow(female)

cond_femsur <- fem_survived/female
cat("The probability of surviving given you are a female is for:",
cond_femsur*100, "%")

## The probability of surviving given you are a female is for: 74.20382 %

#CALCULAR PROBABILIDAD CONDICIONAL DE SOREVIRIR DADO QUE ES HOMBRE

male <- subset(cleaninfo, Sex == "male")
male_survived <- nrow(subset(male, Survived == 1))
male <- nrow(male)

cond_malesur <- male_survived/male
cat("The probability of surviving given you are a male is for:",
cond_malesur*100, "%")

## The probability of surviving given you are a male is for: 18.89081 %

```

#INTERPRETACIÓN (RESULTADOS) Los datos muestran parcialmente que es más probable sobrevivir siendo mujer que siendo hombre, más sin embargo, el porcentaje de mujeres que sobrevivieron es inferior al porcentaje de hombres que lograron sobrevivir.